

Lasso Regression [L2 regularization]

Lasso (Least absolute Shrinkage and Selection Operator) is a regularized linear regression technique that:

- reduces overfitting, dimensionality
- Handles multicollinearity
- Automatically performs feature selection.

Why do we need Lasso :-

In High-dimensional (nD) data:

- Too many features, some features are useless
- OLS coefficients become unstable.

→ Lasso solves this by forcing unimportant coefficients to zero.

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda ||w||$$

[where $\lambda \geq 0$]

$$\lambda [|w_1| + |w_2| + |w_3| + \dots + |w_n|]$$

When to use lasso :-

→ automatically does feature selection :- irrelevant or weak predictors get coefficient 0 and are dropped, giving a simpler, more interpretable.

→ when you have many features but only a subset are truly important.

→ when you want both good prediction and built in feature selection (high-dimensional data, many correlated or noisy features)

How Coefficients get affected in Lasso:-

• Shrink coefficients

- all coefficients are pulled to zero
- stronger λ = more shrinkage

Eliminates weak features

Small coefficients → set to zero

large coefficients → reduced but kept

Geometric Intuition:-

→ loss constraint $|w_1| + |w_2| \leq C$

- Shape diamond
- Error contours are ellipses
- ellipses often touch ^{short} corner of diamond
- corner lies on axis
- axis $\Rightarrow$ one coefficient becomes exactly 0

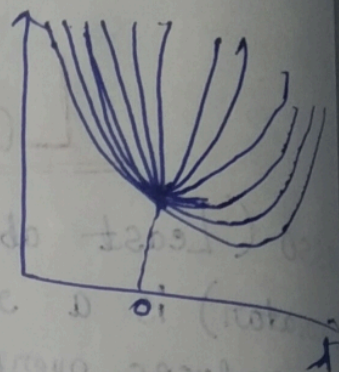
→ This is why lasso does feature selection

$$\lambda |w|$$

$\Rightarrow$ Large $|w| \Rightarrow$ larger penalty

→ But penalty grows linearly not quadratically

Geometric Intuition



Why Lasso Creates Sparsity:-

Sparsity means:-

- many coefficients become exactly zero
- only a few features are kept

Simple:-

$$X|Y \Rightarrow y = mx + b \Rightarrow m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

from Ridge:-

$$b = \bar{y} - m\bar{x}$$

$\rightarrow \bar{y} \rightarrow \text{mean}(y)$
 $\rightarrow (\bar{x}) \rightarrow \text{mean}(x)$

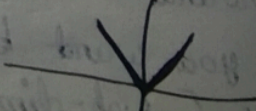
$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

Now for Lasso Regression

$$b = \bar{y} - m\bar{x}, m = ?$$

→ add penalty of $\lambda |m|$

when $m = 0, |m|$ is not differential at 0



So, when $m > 0$ $L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda |m|$

$$\frac{dL}{dm} = \sum_{i=1}^n (y_i - mx_i - \frac{\bar{y} + m\bar{x}}{b})^2 + 2\lambda m \quad (m > 0) \quad \text{So } (x_2)$$

$$\Rightarrow \sum_{i=1}^n 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda = 0$$

$$\Rightarrow -\sum_{i=1}^n [(y_i - \bar{y}) - m(x_i - \bar{x})](x_i - \bar{x}) + \lambda = 0$$

$$\Rightarrow -\sum_{i=1}^n [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] + \lambda = 0$$

$$\Rightarrow -\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda = 0$$

$$m \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - \lambda = 0$$

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\lambda |m| \Rightarrow -m$$

i) for $m > 0$

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

ii) for $m = 0$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

iii) for $m < 0$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum (x_i - \bar{x})^2}$$

case i: $m > 0$ | but always $\lambda > 0$

Example:- $\frac{yX - \lambda}{x^2} \begin{cases} yX = 100 \\ x^2 = 50 \end{cases}$

$\lambda = 0$ $m = 2$	$\lambda = 10$ $m = 9/5$	$\lambda = 100$ $m = 0$	$\lambda \geq 100$ $m = -1$	$\lambda = 150$ $m = -5$
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here $m < 0$, now it will apply this

→ our goal is to reduce loss, 2, 9/5, 1, 0, -1, -5

→ after when $\lambda < 0$, it starts increases, so we are stopping it at zero.

Case 2: $m < 0$, Consider $m = \frac{-100 + \lambda}{50}$

when $\lambda = 0 \rightarrow m = -2$

$\lambda = 50 \rightarrow m = -1$

$\lambda = 100 \rightarrow m = 0$

$\lambda \geq 150$

now $m > 0$, Consider case 1

$\lambda = 150 \rightarrow m = \frac{-100 - 150}{50}$
 $m = -5$

→ so it's start increasing again, when we reach zero we will stop at zero

Why Ridge Doesn't Create Sparsity:

for ridge:-

$$m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

Ridge $\lambda > 0$

→ here λ is in the denominator

→ Even if $(y_i - \bar{y})(x_i - \bar{x}) = 0.1$, & $\lambda = 10000$ we will get very small numbers, which is

close to zero, but not exactly zero.

→ generally numerator will not become zero

→ but in Lasso :- we have λ in numerator itself & since it's an hyperparameter, there is a chance to make numerator zero.

mathematical view:- no corners due to $(\beta)^2$

Ridge closed-form $\Rightarrow \beta = (X^T X + \lambda I)^{-1} X^T y$

→ matrix inverse → Coefficients never become exactly zero only become very small. close to zero but not exactly zero. $\beta^2 \Rightarrow$ smooth everywhere, derivative is 0 is 0. no sharp corner.

Lasso :-
• No closed form / gradient
• uses thresholding
• Coefficients 'Snap to zero'

$|\beta|$
→ Gradient is undefined at 0
→ pull towards zero

$$\lambda |\beta_1| + \lambda |\beta_2|$$

